

GOODLAND, KS, USA

WMO: 724650

Lat: 39.367N

Lon: 101.693W

Elev: 1114

StdP: 88.63

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 23065

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.3	-14.3	-21.1	0.7	-14.9	-18.5	0.8	-11.0	15.8	3.9	14.2	3.8	4.9	340	0.684

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.2	36.5	19.2	34.9	19.3	33.3	19.2	21.9	30.5	21.3	30.0	20.6	29.3	5.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.6	16.5	24.6	18.9	15.7	24.0	18.1	15.0	23.4	69.8	30.5	67.0	29.8	64.5	29.3	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
14.3	12.1	10.8		-22.2	38.9	2.5	1.5	-24.0	39.9	-25.5	40.8	-26.9	41.7	-28.8	42.7	
				-22.5	23.3	2.5	0.9	-24.3	23.9	-25.7	24.4	-27.1	24.9	-28.9	25.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.1	-0.8	0.2	5.2	9.6	15.3	21.5	24.6	23.3	18.6	11.2	4.6	-0.6
	DBStd	10.46	6.17	6.51	6.08	5.38	4.86	3.99	3.09	2.98	4.75	5.41	5.69	5.92
	HDD10.0	1418	335	277	168	71	11	1	0	0	3	49	176	328
	HDD18.3	3186	593	508	406	263	119	16	2	3	53	226	412	586
	CDD10.0	1835	1	3	21	59	174	346	454	413	262	88	14	1
	CDD18.3	559	0	0	0	2	24	112	197	157	62	6	0	0
	CDH23.3	7121	1	3	32	109	436	1426	2371	1731	839	163	9	0
	CDH26.7	3471	0	1	5	26	161	694	1298	864	373	49	0	0
Wind	WSAvg	5.3	5.2	5.5	5.9	6.3	5.7	5.4	4.8	4.6	5.0	5.3	5.2	5.2
Precipitation	PrecAvg	468	10	11	26	38	78	81	69	56	36	31	16	12
	PrecMax	752	40	41	91	123	208	240	202	153	137	109	54	71
	PrecMin	234	0	0	0	0	12	2	10	3	0	0	0	0
	PrecStd	111	8	10	22	26	45	52	40	40	32	29	14	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.6	22.3	27.0	29.6	33.7	38.2	38.5	37.4	35.5	31.2	24.7	18.9
		MCWB	7.4	8.7	11.6	12.9	16.5	18.2	19.5	19.1	17.5	14.5	10.5	8.0
	2%	DB	15.8	17.8	23.3	26.6	30.8	35.2	36.9	35.2	33.2	27.7	21.5	15.2
		MCWB	6.0	7.1	9.8	12.4	15.7	18.6	19.7	19.5	17.6	13.6	9.5	6.0
	5%	DB	12.1	14.3	20.0	23.7	28.1	33.1	35.3	33.6	31.0	24.6	18.1	11.8
		MCWB	4.4	5.7	8.9	11.7	15.7	18.6	19.9	19.5	17.4	13.0	8.4	4.4
	10%	DB	8.7	10.5	16.6	20.5	25.5	30.9	33.6	31.8	28.5	21.3	14.4	8.6
		MCWB	2.8	3.7	7.7	10.8	15.0	18.5	20.0	19.5	16.6	11.7	6.8	3.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.3	9.7	13.3	15.1	19.5	22.1	23.0	22.7	20.6	17.1	12.2	8.5
		MCDB	17.9	19.7	22.6	23.8	27.9	30.3	31.3	30.8	29.3	24.8	20.2	17.1
	2%	WB	6.3	7.5	11.3	13.8	18.0	20.9	22.2	21.8	19.6	15.4	10.4	6.4
		MCDB	15.3	16.9	20.0	22.5	26.0	29.4	31.0	30.0	28.5	22.7	19.6	14.2
	5%	WB	4.8	5.9	9.6	12.6	16.9	20.1	21.5	21.0	18.7	14.1	8.9	4.8
		MCDB	11.7	13.6	18.8	21.3	25.0	29.1	30.5	29.3	27.0	22.3	17.3	11.6
	10%	WB	3.0	4.1	8.0	11.3	15.7	19.3	20.8	20.4	17.8	12.7	7.2	2.9
		MCDB	8.4	10.2	16.1	19.6	23.5	28.1	29.9	28.6	25.7	19.8	14.2	8.3
Mean Daily Temperature Range	5% DB	MDBR	14.0	14.0	15.7	15.2	14.7	15.3	15.2	14.6	15.5	15.3	14.8	13.8
		MCDBR	19.0	19.7	21.3	21.1	18.9	18.7	17.9	17.5	18.5	20.0	20.3	18.8
		MCWBR	11.1	11.1	10.7	9.8	7.7	6.3	5.4	5.5	6.4	8.8	10.6	11.2
	5% WB	MCDBR	18.0	18.8	19.5	18.1	16.4	16.1	15.5	14.8	16.1	17.0	19.1	18.0
		MCWBR	10.9	11.0	10.5	9.2	7.7	6.5	5.6	5.7	6.4	8.4	10.3	11.0
Clear-Sky Solar Irradiance	taub	0.251	0.259	0.286	0.314	0.339	0.357	0.375	0.375	0.358	0.328	0.288	0.259	0.250
	taud	2.568	2.559	2.490	2.436	2.432	2.412	2.412	2.407	2.417	2.490	2.575	2.579	2.561
	Ebn at Noon	948	982	978	963	938	918	897	906	919	933	929	923	
	Edh at Noon	70	82	98	111	114	116	116	112	97	80	69	66	
All-Sky Solar Radiation	RadAvg	2.55	3.36	4.64	5.69	6.57	7.35	7.31	6.37	5.22	3.81	2.78	2.21	
	RadStd	0.18	0.26	0.37	0.28	0.40	0.32	0.26	0.31	0.28	0.35	0.19	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page